

Semiconductor Alloy Discovery Report

Materials Project · Bronze → Silver → Gold · DFT-grounded scores · Toxicity & scarcity filters

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Pipeline reproducible con trazabilidad SHA-256, quality gates, SQL de limpieza, JOINS muchos-a-muchos, EXPLAIN ANALYZE y star-schema Gold. Scores calculados a partir de propiedades DFT reales de Materials Project. Se aplican tres filtros nuevos: **(1) exclusión de elementos tóxicos/radiactivos** (Hg, Be, Tl, Pb, Th, U...), **(2) exclusión de tierras raras y metales escasos** (Ho, Er, Os, Ir, Re...), **(3) penalización del 10 % a materiales puramente teóricos**.

1. Metodología de Scoring (DFT real)

Métrica	Campo API	Fórmula	Peso	Referencia física
band_gap_score_sq	band_gap	Gauss(Eg,1.34,0.4)	25%	Límite S-Q: óptimo 1.34 eV
band_gap_score_si	band_gap	Gauss(Eg,1.12,0.3)	15%	Referencia Si: 1.12 eV
stability_score	energy_above_hull	exp(-E_hull/0.05)	25%	E_hull=0 → casco convexo
dielectric_score	n (DFT)	Gauss(n,3.25,0.75)	15%	n∈ [2.5,4.0] → optoelectrónica
mechanical_score	bulk_modulus_vrh	Gauss(B,130,60)	10%	B~80–200 GPa (Si:98,GaAs:75)
magnetic_score	ordering	NM=1 AFM=0.5 FM=0	10%	No magnético preferido
direct_gap_bonus	is_gap_direct	+0.10 si directa	bonus	Emisión eficiente (LEDs)
theoretical_penalty	theoretical	×0.90 si teórico	—	No sintetizado aún

Tabla 1. Scores y filtros del pipeline.

2. Top-20 Candidatos Semiconductores

Excluidos: tóxicos, escasos. CSV: [data/gold/top_semiconductor_candidates.csv](#)

ID	Fórmula	E _g (eV)	Gap	n	ε _{tot}	B (GPa)	E _{hull}	ΔH _f	Sist.	n _{el}	Est.	Teór.	Score
mp-4763	ZnSiP2	1.364	Dir.	3.24	12.4	85	0.0000	-0.392	Tetragonal	3.0	Sí	No	1.0327
mp-1016197	MgSiAs2	1.303	Dir.	3.19	12.4	58	0.0000	-0.297	Tetragonal	3.0	Sí	No	1.0216
mp-1078908	Ca2CdP2	1.293	Dir.	3.06	18.3	-	0.0000	-0.893	Orthorhombic	3.0	Sí	No	1.0204
mp-1205337	Ca3(AlP2)2	1.248	Dir.	2.95	18.9	-	0.0000	-1.176	Monoclinic	3.0	Sí	No	1.0186
mp-1105549	BaHfS3	1.252	Dir.	2.93	58.2	-	0.0000	-2.140	Orthorhombic	3.0	Sí	No	1.0167
mp-1189305	Sr3(AlAs2)2	1.146	Dir.	3.05	18.1	-	0.0000	-1.155	Monoclinic	3.0	Sí	No	1.0165
mp-19913	Ba3(InP2)2	1.109	Dir.	3.12	21.8	-	0.0000	-0.786	Monoclinic	3.0	Sí	No	1.0092
mp-29163	Ag2TeS3	1.300	Dir.	2.89	13.1	-	0.0000	-0.079	Monoclinic	3.0	Sí	No	1.0076
mp-13275	NaSrP	1.236	Dir.	2.80	18.5	-	0.0000	-0.964	Hexagonal	3.0	Sí	No	1.0067
mp-14206	K3Ag3As2	1.253	Dir.	2.81	14.6	-	0.0000	-0.350	Trigonal	3.0	Sí	No	1.0064
mp-2961	MgSiP2	1.373	Dir.	2.94	11.6	72	0.0000	-0.411	Tetragonal	3.0	Sí	No	1.0046
mp-4666	CdSiP2	1.426	Dir.	3.29	13.5	74	0.0000	-0.273	Tetragonal	3.0	Sí	No	0.9985
mp-27175	InOF	1.331	Dir.	-	-	135	0.0000	-2.376	Orthorhombic	3.0	Sí	No	0.9918
mp-9843	Sr3(AlP2)2	1.337	Dir.	2.87	16.9	51	0.0000	-1.163	Monoclinic	3.0	Sí	No	0.9896
mp-1190	ZnSe	1.166	Dir.	2.71	10.7	56	0.0000	-0.952	Cubic	2.0	Sí	No	0.9882
mp-14001	Ba4SiAs4	1.036	Dir.	3.15	21.3	-	0.0000	-0.995	Cubic	3.0	Sí	No	0.9804
mp-28347	K3Al2As3	1.186	Dir.	2.59	12.8	-	0.0000	-0.558	Monoclinic	3.0	Sí	No	0.9804
mp-2176	ZnTe	1.073	Dir.	3.00	12.0	43	0.0000	-0.684	Cubic	2.0	Sí	No	0.9752
mp-571195	ZnTe	1.069	Dir.	3.03	12.2	-	0.0037	-0.680	Trigonal	2.0	No	No	0.9727
mp-9666	K2NaGaP2	1.308	Dir.	2.57	13.1	-	0.0000	-0.443	Orthorhombic	4.0	Sí	No	0.9719

Tabla 2. Top-20 candidatos — sin tóxicos ni escasos.

3. Top-30 Candidatos a Aleaciones (nelem ≥ 2)

Todos los atributos de evaluación. CSV: [data/gold/alloy_candidates.csv](#)

ID	Fórmula	n_el	Eg	Gap	Sc_SQ	Sc_Si	E_hull	Sc_Est	n	ϵ_e	Sc_Di	B	Sc_Me	Mag	Sc_Ma	Bonus	Score
mp-4763	ZnSiP2	3.0	1.364	Dir.	0.998	0.718	0.0000	1.000	3.24	10.53	1.000	85	0.754	No	1.00	0.10	1.0327
mp-1016197	MgSiAs2	3.0	1.303	Dir.	0.996	0.830	0.0000	1.000	3.19	10.17	0.997	58	0.487	No	1.00	0.10	1.0216
mp-1078908	Ca2CdP2	3.0	1.293	Dir.	0.993	0.847	0.0000	1.000	3.06	9.34	0.967	-	0.500	No	1.00	0.10	1.0204
mp-1205337	Ca3(AlP2)2	3.0	1.248	Dir.	0.974	0.913	0.0000	1.000	2.95	8.68	0.921	-	0.500	No	1.00	0.10	1.0186
mp-1105549	BaHfS3	3.0	1.252	Dir.	0.976	0.907	0.0000	1.000	2.93	8.56	0.910	-	0.500	No	1.00	0.10	1.0167
mp-1189305	Sr3(AlAs2)2	3.0	1.146	Dir.	0.889	0.996	0.0000	1.000	3.05	9.32	0.966	-	0.500	No	1.00	0.10	1.0165
mp-19913	Ba3(InP2)2	3.0	1.109	Dir.	0.847	0.999	0.0000	1.000	3.12	9.71	0.984	-	0.500	No	1.00	0.10	1.0092
mp-29163	Ag2TeS3	3.0	1.300	Dir.	0.995	0.835	0.0000	1.000	2.89	8.34	0.890	-	0.500	No	1.00	0.10	1.0076
mp-13275	NaSrP	3.0	1.236	Dir.	0.967	0.928	0.0000	1.000	2.80	7.87	0.839	-	0.500	No	1.00	0.10	1.0067
mp-14206	K3Ag3As2	3.0	1.253	Dir.	0.977	0.906	0.0000	1.000	2.81	7.90	0.842	-	0.500	No	1.00	0.10	1.0064
mp-2961	MgSiP2	3.0	1.373	Dir.	0.997	0.701	0.0000	1.000	2.94	8.62	0.916	72	0.629	No	1.00	0.10	1.0046
mp-4666	CdSiP2	3.0	1.426	Dir.	0.977	0.595	0.0000	1.000	3.29	10.82	0.999	74	0.651	No	1.00	0.10	0.9985
mp-27175	InOF	3.0	1.331	Dir.	1.000	0.781	0.0000	1.000	-	-	0.500	135	0.996	No	1.00	0.10	0.9918
mp-9843	Sr3(AlP2)2	3.0	1.337	Dir.	1.000	0.771	0.0000	1.000	2.87	8.23	0.878	51	0.422	No	1.00	0.10	0.9896
mp-1190	ZnSe	2.0	1.166	Dir.	0.910	0.988	0.0000	1.000	2.71	7.32	0.769	56	0.472	No	1.00	0.10	0.9882
mp-14001	Ba4SiAs4	3.0	1.036	Dir.	0.749	0.962	0.0000	1.000	3.15	9.94	0.992	-	0.500	No	1.00	0.10	0.9804
mp-28347	K3Al2As3	3.0	1.186	Dir.	0.929	0.976	0.0000	1.000	2.59	6.71	0.679	-	0.500	No	1.00	0.10	0.9804
mp-2176	ZnTe	2.0	1.073	Dir.	0.801	0.988	0.0000	1.000	3.00	9.02	0.947	43	0.348	No	1.00	0.10	0.9752
mp-571195	ZnTe	2.0	1.069	Dir.	0.795	0.986	0.0037	0.929	3.03	9.20	0.959	-	0.500	No	1.00	0.10	0.9727
mp-9666	K2NaGaP2	4.0	1.308	Dir.	0.997	0.822	0.0000	1.000	2.57	6.60	0.663	-	0.500	No	1.00	0.10	0.9719
mp-14091	AlAgSe2	3.0	1.135	Dir.	0.877	0.999	0.0000	1.000	2.65	7.05	0.729	52	0.433	No	1.00	0.10	0.9719
mp-1101989	SrLiAs	3.0	1.032	Dir.	0.744	0.958	0.0000	1.000	2.99	8.94	0.941	-	0.500	No	1.00	0.10	0.9709
mp-2412	LiP5	2.0	1.367	Dir.	0.998	0.712	0.0077	0.857	3.25	10.59	1.000	-	0.500	No	1.00	0.10	0.9705
mp-380	ZnSe	2.0	1.200	Dir.	0.941	0.965	0.0040	0.923	2.68	7.18	0.748	56	0.470	No	1.00	0.10	0.9699
mp-3917	Cd2GeO4	3.0	1.092	Dir.	0.826	0.996	0.0000	1.000	-	-	0.500	100	0.881	No	1.00	0.10	0.9688

ID	Fórmula	n_el	Eg	Gap	Sc_SQ	Sc_Si	E_hull	Sc_Est	n	ϵ_e	Sc_Di	B	Sc_Me	Mag	Sc_Ma	Bonus	Score
mp-10577	Sr5NbN5	3.0	1.247	Dir.	0.973	0.915	0.0000	1.000	2.48	6.14	0.588	-	0.500	No	1.00	0.10	0.9687
mp-5475	NaTaN2	3.0	1.276	Ind.	0.987	0.873	0.0000	1.000	2.99	8.91	0.940	136	0.995	No	1.00	0.00	0.9682
mp-1080438	SrAgTeF	4.0	1.285	Dir.	0.990	0.860	0.0000	1.000	2.50	6.27	0.610	-	0.500	No	1.00	0.10	0.9681
mp-23202	InI	2.0	1.341	Dir.	1.000	0.762	0.0000	1.000	2.91	8.46	0.901	13	0.151	No	1.00	0.10	0.9646
mp-540771	BaZrS3	3.0	0.990	Dir.	0.682	0.910	0.0000	1.000	3.10	9.59	0.979	70	0.606	No	1.00	0.10	0.9646

Tabla 3. Top-30 aleaciones candidatas con todos los atributos DFT.

4. Mejores Aleaciones — Criterios Estrictos

nelements $\geq$ 2 · E_hull<0.05 · is_stable · $\neg$ magnético · 0.5 $\leq$ Eg $\leq$ 3.5 · $\neg$ tóxico · $\neg$ escaso. CSV: [data/gold/best_semiconductor_alloys.csv](#)

4a. Propiedades físicas DFT:

ID	Fórmula	n_el	Elementos	Sist.	Eg (eV)	Gap	n (DFT)	ϵ_{tot}	B (GPa)	G (GPa)	E_hull	$\Delta H_f/\text{át}$	Est.	Teór.
mp-4763	ZnSiP2	3	P,Si,Zn	Tetrago	1.364	Dir.	3.24	12.4	85	54	0.0000	-0.392	Sí	No
mp-1016197	MgSiAs2	3	As,Mg,Si	Tetrago	1.303	Dir.	3.19	12.4	58	29	0.0000	-0.297	Sí	No
mp-1078908	Ca2CdP2	3	Ca,Cd,P	Orthorh	1.293	Dir.	3.06	18.3	-	-	0.0000	-0.893	Sí	No
mp-1205337	Ca3(AlP2)2	3	Al,Ca,P	Monocli	1.248	Dir.	2.95	18.9	-	-	0.0000	-1.176	Sí	No
mp-1105549	BaHfS3	3	Ba,Hf,S	Orthorh	1.252	Dir.	2.93	58.2	-	-	0.0000	-2.140	Sí	No
mp-1189305	Sr3(AlAs2)2	3	Al,As,Sr	Monocli	1.146	Dir.	3.05	18.1	-	-	0.0000	-1.155	Sí	No
mp-19913	Ba3(InP2)2	3	Ba,In,P	Monocli	1.109	Dir.	3.12	21.8	-	-	0.0000	-0.786	Sí	No
mp-29163	Ag2TeS3	3	Ag,S,Te	Monocli	1.300	Dir.	2.89	13.1	-	-	0.0000	-0.079	Sí	No
mp-13275	NaSrP	3	Na,P,Sr	Hexagon	1.236	Dir.	2.80	18.5	-	-	0.0000	-0.964	Sí	No
mp-14206	K3Ag3As2	3	Ag,As,K	Trigona	1.253	Dir.	2.81	14.6	-	-	0.0000	-0.350	Sí	No
mp-2961	MgSiP2	3	Mg,P,Si	Tetrago	1.373	Dir.	2.94	11.6	72	35	0.0000	-0.411	Sí	No
mp-4666	CdSiP2	3	Cd,P,Si	Tetrago	1.426	Dir.	3.29	13.5	74	40	0.0000	-0.273	Sí	No
mp-27175	InOF	3	F,In,O	Orthorh	1.331	Dir.	-	-	135	57	0.0000	-2.376	Sí	No
mp-9843	Sr3(AlP2)2	3	Al,P,Sr	Monocli	1.337	Dir.	2.87	16.9	51	30	0.0000	-1.163	Sí	No
mp-1190	ZnSe	2	Se,Zn	Cubic	1.166	Dir.	2.71	10.7	56	27	0.0000	-0.952	Sí	No
mp-14001	Ba4SiAs4	3	As,Ba,Si	Cubic	1.036	Dir.	3.15	21.3	-	-	0.0000	-0.995	Sí	No
mp-28347	K3Al2As3	3	Al,As,K	Monocli	1.186	Dir.	2.59	12.8	-	-	0.0000	-0.558	Sí	No
mp-2176	ZnTe	2	Te,Zn	Cubic	1.073	Dir.	3.00	12.0	43	23	0.0000	-0.684	Sí	No
mp-9666	K2NaGaP2	4	Ga,K,Na,P	Orthorh	1.308	Dir.	2.57	13.1	-	-	0.0000	-0.443	Sí	No
mp-14091	AlAgSe2	3	Ag,Al,Se	Tetrago	1.135	Dir.	2.65	9.7	52	19	0.0000	-0.743	Sí	No
mp-1101989	SrLiAs	3	As,Li,Sr	Orthorh	1.032	Dir.	2.99	19.8	-	-	0.0000	-1.200	Sí	No
mp-3917	Cd2GeO4	3	Cd,Ge,O	Orthorh	1.092	Dir.	-	-	100	34	0.0000	-1.762	Sí	No
mp-10577	Sr5NbN5	3	N,Nb,Sr	Orthorh	1.247	Dir.	2.48	17.8	-	-	0.0000	-1.042	Sí	No

ID	Fórmula	n_el	Elementos	Sist.	Eg (eV)	Gap	n (DFT)	ε_tot	B (GPa)	G (GPa)	E_hull	ΔHf/át	Est.	Teór.
mp-5475	NaTaN2	3	N,Na,Ta	Trigona	1.276	Ind.	2.99	18.3	136	89	0.0000	-1.139	Sí	No
mp-1080438	SrAgTeF	4	Ag,F,Sr,Te	Tetrago	1.285	Dir.	2.50	15.6	-	-	0.0000	-2.160	Sí	No
mp-23202	InI	2	I,In	Orthorh	1.341	Dir.	2.91	43.3	13	7	0.0000	-0.652	Sí	No
mp-540771	BaZrS3	3	Ba,S,Zr	Orthorh	0.990	Dir.	3.10	110.4	70	34	0.0000	-2.108	Sí	No
mp-8017	AlCuTe2	3	Al,Cu,Te	Tetrago	1.019	Dir.	3.17	11.7	50	25	0.0000	-0.425	Sí	No
mp-17696	Ba2VN3	3	Ba,N,V	Orthorh	1.388	Dir.	2.69	21.5	-	-	0.0000	-1.217	Sí	No
mp-14092	AlAgTe2	3	Ag,Al,Te	Tetrago	1.047	Dir.	2.97	11.0	42	17	0.0000	-0.439	Sí	No

Tabla 4a. Propiedades físicas — top-30 mejores aleaciones (100 totales cumplen criterios estrictos).

4b. Desglose de scores:

Fórmula	Sc_SQ	Sc_Si	Sc_Est.	Sc_Diel.	Sc_Mec.	Sc_Mag.	Bonus_Dir.	Score Total
ZnSiP2	0.998	0.718	1.000	1.000	0.754	1.00	0.10	1.0327
MgSiAs2	0.996	0.830	1.000	0.997	0.487	1.00	0.10	1.0216
Ca2CdP2	0.993	0.847	1.000	0.967	0.500	1.00	0.10	1.0204
Ca3(AlP2)2	0.974	0.913	1.000	0.921	0.500	1.00	0.10	1.0186
BaHfS3	0.976	0.907	1.000	0.910	0.500	1.00	0.10	1.0167
Sr3(AlAs2)2	0.889	0.996	1.000	0.966	0.500	1.00	0.10	1.0165
Ba3(InP2)2	0.847	0.999	1.000	0.984	0.500	1.00	0.10	1.0092
Ag2TeS3	0.995	0.835	1.000	0.890	0.500	1.00	0.10	1.0076
NaSrP	0.967	0.928	1.000	0.839	0.500	1.00	0.10	1.0067
K3Ag3As2	0.977	0.906	1.000	0.842	0.500	1.00	0.10	1.0064
MgSiP2	0.997	0.701	1.000	0.916	0.629	1.00	0.10	1.0046
CdSiP2	0.977	0.595	1.000	0.999	0.651	1.00	0.10	0.9985
InOF	1.000	0.781	1.000	0.500	0.996	1.00	0.10	0.9918
Sr3(AlP2)2	1.000	0.771	1.000	0.878	0.422	1.00	0.10	0.9896
ZnSe	0.910	0.988	1.000	0.769	0.472	1.00	0.10	0.9882
Ba4SiAs4	0.749	0.962	1.000	0.992	0.500	1.00	0.10	0.9804

Fórmula	Sc_SQ	Sc_Si	Sc_Est.	Sc_Diel.	Sc_Mec.	Sc_Mag.	Bonus_Dir.	Score Total
K3Al2As3	0.929	0.976	1.000	0.679	0.500	1.00	0.10	0.9804
ZnTe	0.801	0.988	1.000	0.947	0.348	1.00	0.10	0.9752
K2NaGaP2	0.997	0.822	1.000	0.663	0.500	1.00	0.10	0.9719
AlAgSe2	0.877	0.999	1.000	0.729	0.433	1.00	0.10	0.9719
SrLiAs	0.744	0.958	1.000	0.941	0.500	1.00	0.10	0.9709
Cd2GeO4	0.826	0.996	1.000	0.500	0.881	1.00	0.10	0.9688
Sr5NbN5	0.973	0.915	1.000	0.588	0.500	1.00	0.10	0.9687
NaTaN2	0.987	0.873	1.000	0.940	0.995	1.00	0.00	0.9682
SrAgTeF	0.990	0.860	1.000	0.610	0.500	1.00	0.10	0.9681
InI	1.000	0.762	1.000	0.901	0.151	1.00	0.10	0.9646
BaZrS3	0.682	0.910	1.000	0.979	0.606	1.00	0.10	0.9646
AlCuTe2	0.724	0.944	1.000	0.994	0.410	1.00	0.10	0.9629
Ba2VN3	0.993	0.672	1.000	0.758	0.500	1.00	0.10	0.9627
AlAgTe2	0.765	0.971	1.000	0.933	0.343	1.00	0.10	0.9610

Tabla 4b. Scores por componente — fila dorada = mejor aleación.

5. Análisis por Elemento (JOIN M:N)

Un elemento por fila. CSV: *data/gold/elements_analysis.csv*

Elemento	Nº mat.	Eg med.	E_hull med.	n med.	B med.(GPa)	Sc_Di	Sc_Me	Score med.	Estables	Dir.	No mag.	Tóxicos	Escasos
Si	13	1.200	0.0001	3.05	72	0.765	0.540	0.9631	12	13	13	0	0
Zn	11	1.209	0.0009	3.04	55	0.834	0.479	0.9617	7	9	11	0	0
Mg	5	1.248	0.0007	3.05	59	0.773	0.502	0.9616	4	4	5	0	0
V	2	1.267	0.0000	2.69	-	0.629	0.500	0.9548	2	2	2	0	0
B	2	1.288	0.0000	3.03	162	0.729	0.684	0.9513	2	1	2	0	0
Sr	20	1.203	0.0008	2.76	60	0.642	0.501	0.9510	16	20	20	0	0
In	10	1.245	0.0005	2.90	57	0.652	0.485	0.9500	9	9	10	0	0
Hf	5	1.322	0.0000	2.90	104	0.807	0.583	0.9495	5	4	5	0	0
Al	18	1.208	0.0007	2.94	61	0.756	0.478	0.9491	16	15	18	0	0
Te	18	1.166	0.0008	2.94	31	0.778	0.395	0.9488	15	17	18	0	0
Ta	6	1.183	0.0008	2.99	136	0.573	0.583	0.9484	5	5	6	0	0
P	43	1.234	0.0003	2.89	78	0.671	0.519	0.9484	39	38	43	0	0
Np	2	1.156	0.0000	-	-	0.500	0.500	0.9466	2	2	2	0	0
F	10	1.246	0.0009	2.53	94	0.551	0.543	0.9457	7	10	10	0	0
C	2	1.236	0.0000	2.86	159	0.686	0.694	0.9446	2	1	2	0	0
Ca	10	1.207	0.0000	2.88	60	0.783	0.501	0.9444	10	7	10	0	0
N	14	1.279	0.0000	2.57	73	0.625	0.535	0.9437	14	12	14	0	0
Cd	15	1.183	0.0004	2.83	59	0.689	0.503	0.9433	14	15	15	0	0
As	26	1.144	0.0002	2.91	54	0.739	0.496	0.9432	25	21	26	0	0
W	5	1.237	0.0000	-	-	0.500	0.500	0.9425	5	5	5	0	0
Na	16	1.218	0.0004	2.68	102	0.671	0.541	0.9421	15	13	16	0	0
K	22	1.233	0.0001	2.59	-	0.555	0.500	0.9421	19	21	22	0	0
Ag	21	1.184	0.0008	2.71	48	0.665	0.480	0.9399	16	19	21	0	0
Nb	9	1.257	0.0004	2.69	152	0.608	0.596	0.9391	8	8	9	0	0
Se	43	1.277	0.0005	2.67	46	0.562	0.479	0.9388	35	41	43	0	0

Tabla 5. Top-25 elementos por score promedio.

6. Resumen Ejecutivo

Indicador	Valor
Materiales en Silver	57299
Con n DFT real	5627
Con módulo volumétrico B (DFT)	2713
Con brecha directa	15693
Flagged tóxicos	4138
Flagged escasos	12528
Candidatos Top-200 (sin tóx/esc.)	200
Candidatos aleaciones Top-200	200
Mejores aleaciones (criterios est.)	100
Score máximo	1.0327
Mejor aleación (fórmula)	ZnSiP2
Eg mejor aleación (eV)	1.364
n mejor aleación	3.24
B_VRH mejor aleación (GPa)	85
E_hull mejor aleación	0.0000

Tabla 6. Resumen ejecutivo.